Radicals & Rational Exponents

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in inverse relationship between exponentials and logarithms, evaluate $f(5)$.

Final Answer: 29. Substitute $x = 5$: $f(5) = 5(5) + 4 = 29$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 5$ in logarithmic functions, evaluate $f(2)$.

Final Answer: 17. Substitute $x = 2$: $f(2) = 6(2) + 5 = 17$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for logarithmic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 24.5. The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 6$ in extension: properties of logarithms, evaluate $f(2)$.

Final Answer: 10. Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for extension: properties of logarithms consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 16.5. The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 3x + 7$ in exponential and logarithmic equations, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 3(2) + 7 = 13$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for exponential and logarithmic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 17.5. The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the equation $\sqrt{2x+7} = x+2$ for all real values of x.

Square both sides of the equation: $(\sqrt{2x+7})^2 = (x+2)^2 \Rightarrow 2x+7 = x^2+4x+4$. Rearrange into standard quadratic form: $x^2+2x-3=0 \Rightarrow (x+3)(x-1)=0 \Rightarrow x=1$ or $x=-3$. Now check both candidate roots in the original equation:For $x=1$: LHS = $\sqrt{2(1)+7} = \sqrt{9} = 3$; RHS = $1+2=3$. (Valid ✓)For $x=-3$: LHS = $\sqrt{2(-3)+7} = \sqrt{1} = 1$; RHS = $-3+2=-1$. Since $1eq-1$, $x=-3$ is extraneous.The only real solution is.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Simplify and rationalize the denominator: $\frac{\sqrt{3}}{\sqrt{5}-\sqrt{2}}$.

Multiply numerator and denominator by the conjugate $\sqrt{5}+\sqrt{2}$: $\frac{\sqrt{3}(\sqrt{5}+\sqrt{2})}{(\sqrt{5}-\sqrt{2})(\sqrt{5}+\sqrt{2})}$. Expand denominator using difference of squares $(a-b)(a+b)=a^2-b^2$: $(\sqrt{5})^2-(\sqrt{2})^2 = 5-2=3$. Expand numerator and simplify: $\frac{\sqrt{15}+\sqrt{6}}{3}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Radicals & Exponents · Practice 3] If the governing relation for this system is $g(t) = 5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Substitute $t=8$: $g(8) = 5(8)+5 = \mathbf{45}$. Setting $5t+5=50$ gives $5t=45 \Rightarrow t=\mathbf{9}$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Evaluate the numerical value of $16^{-3/4} + \left(\frac{27}{8}\right)^{-2/3}$.

1. Evaluate the first term: $16^{-3/4} = \frac{1}{16^{3/4}} = \frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3} = \frac{1}{8}$. 2. Evaluate the second term: inverting the fraction removes the negative exponent: $\left(\frac{27}{8}\right)^{-2/3} = \left(\frac{8}{27}\right)^{2/3} = \frac{(\sqrt[3]{8})^2}{(\sqrt[3]{27})^2} = \frac{2^2}{3^2} = \frac{4}{9}$. 3. Combine with a common denominator of 72: $\frac{1}{8}+\frac{4}{9} = \frac{9}{72}+\frac{32}{72} = \frac{41}{72}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Simplify the algebraic expression $\frac{(2x^3y^{-2})^3}{(4x^{-1}y^4)^2}$ leaving only positive exponents.

Apply power of a power to numerator and denominator separately:
Numerator = $2^3 \cdot (x^3)^3 \cdot (y^{-2})^3 = 8x^9y^{-6}$. Denominator = $4^2 \cdot (x^{-1})^2 \cdot (y^4)^2 = 16x^{-2}y^8$. Divide coefficients and subtract exponents: $\frac{8}{16} \cdot x^{9-(-2)} \cdot y^{-6-8} = \frac{1}{2}x^{11}y^{-14} = \frac{x^{11}}{2y^{14}}$.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The ACT Subject Test in Mathematics consists of 50 questions in 60 minutes. If a student spends an average of 55 seconds on each of the first 30 questions, how many seconds per question are available for the remaining 20 questions?

Time spent on first 30 questions = $30 \times 55 = 1650$ seconds (27.5 minutes).
Total time available = $60 \times 60 = 3600$ seconds. Remaining time = $3600 - 1650 = 1950$ seconds. For the final 20 questions: $\frac{1950}{20} = 97.5$ seconds per question.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · MediumA table of values for a positive function $g(x)$ over equally spaced inputs $x = 0, 1, 2, 3$ produces outputs $g(0) = 4, g(1) = 12, g(2) = 36, g(3) = 108$. Which function family best models $g(x)$, and what is the constant growth factor per unit increase in x?

Calculate the consecutive ratios of consecutive outputs: $\frac{g(1)}{g(0)} = \frac{12}{4} = 3, \quad \frac{g(2)}{g(1)} = \frac{36}{12} = 3, \quad \frac{g(3)}{g(2)} = \frac{108}{36} = 3.$ Because the ratio of outputs is constant for equal-length input intervals, $g(x)$ is an exponential function with a constant growth factor of 3.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Matrices & Systems

A set of five exam scores has a mean of 84. Four of the scores are 78, 86, 92, and 80. Find the 5th score, and find the new mean if the lowest score is removed.

Total sum of the 5 scores = $5 \times 84 = 420$. Sum of the 4 known scores = $78 + 86 + 92 + 80 = 336$. 5th score = $420 - 336 = 84$. Removing the lowest score (78) leaves four scores: 80, 84, 86, 92. New sum = $420 - 78 = 342$. New mean = $342/4 = 85.5$ (an increase of 1.5 points).

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Probability & Statistics

[AP Precalculus Exam Format & Modeling Framework · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

[Unit 1 · Polynomial and Rational Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Polynomial Functions

[Exponents & Polynomials · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Rational Expressions

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a = 7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{9^2 + 10^2 - 7^2}{2(9)(10)} = \frac{81 + 100 - 49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333$. $A = \arccos(11/15) \approx 42.8^\circ$. Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7 + 9 + 10}{2} = 13.$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in graphs and characteristics of exponential functions, evaluate $f(2)$.

a. 15 ✓

b. 18

c. 6

d. 19

Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(2)$.

a. 11 ✓

b. 15

c. 8

d. 15

Substitute $x = 2$: $f(2) = 4(2) + 3 = 11$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for graphs and characteristics of exponential functions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for applications of exponential functions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For a constant k, the equation $\sqrt{2x + k} = x - 1$ has $x = 5$ as a valid solution. What is the extraneous solution produced when solving this equation by squaring both sides?

a. -5

b. -1 ✓

c. 1

d. 6

We solve the problem in two main steps: 1. **Find the parameter k:** Since $x = 5$ is a solution to the original radical equation, substitute $x = 5$ into the equation: $\sqrt{2(5) + k} = 5 - 1$ $\sqrt{10 + k} = 4$ Squaring both sides gives: $10 + k = 16 \Rightarrow k = 6$ 2. **Solve the equation and find the extraneous solution:** Substitute $k = 6$ back into the equation: $\sqrt{2x + 6} = x - 1$ Square both sides to eliminate the radical: $2x + 6 = (x - 1)^2$ $2x + 6 = x^2 - 2x + 1$ Rearrange all terms to one s.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Consider the equation $\sqrt[3]{25 - x^2} + \sqrt[3]{x^2 + 3} = 4$ What is the product of all real solutions to this equation?

a. -24 ✓

b. 24

c. -48

d. 0

To solve the equation, let $u = 25 - x^2$ and $v = x^2 + 3$. Observe that the sum of the radicands eliminates x^2: $u + v = (25 - x^2) + (x^2 + 3) = 28$ The original equation can be written in terms of rational exponents: $u^{1/3} + v^{1/3} = 4$ Let $a = u^{1/3}$ and $b = v^{1/3}$. Then: $a + b = 4$ and $a^3 + b^3 = 28$ Using the algebraic identity $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$: $28 = 4^3 - 3ab(4)$ $28 = 64 - 12ab$ $12ab = 36 \Rightarrow ab = 3$ Since $a + b = 4$ and $ab = 3$, a and.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)
Consider the function $g(x) = \sqrt{x^2 - 6x + 9} + \sqrt{x^2 + 8x + 16}$. Find the minimum possible value of $g(x)$.

a. 2

b. 3

c. 4

d. **7** ✓

First, we simplify the terms under the square roots. The first term is $\sqrt{x^2 - 6x + 9}$. This is a perfect square trinomial, which factors as $(x - 3)^2$. Therefore, $\sqrt{x^2 - 6x + 9} = \sqrt{(x - 3)^2} = |x - 3|$. The second term is $\sqrt{x^2 + 8x + 16}$. This is also a perfect square trinomial, which factors as $(x + 4)^2$. Therefore, $\sqrt{x^2 + 8x + 16} = \sqrt{(x + 4)^2} = |x + 4|$. So, the function can be rewritten as $g(x) = |x - 3| + |x + 4|$. This expression represents the sum of the distances from x to 3 an.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
What is the value of the expression $\sqrt{36} + \sqrt{49}$?

a. 10

b. 11

c. **13** ✓

d. 15

First, calculate the square root of 36. Since $6^2 = 36$, $\sqrt{36} = 6$. Next, calculate the square root of 49. Since $7^2 = 49$, $\sqrt{49} = 7$. Finally, add the two results: $6 + 7 = 13$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)
Check: is $x = -1$ a solution of $\sqrt{x} = -1$?

a. Maybe

b. Only in $\mathbb{C}$

c. **No (square root is non-negative)** ✓

d. Yes

Applying Radicals & Rational Exponents principles, $\sqrt{x} \geq 0$ for real $x \geq 0$. Therefore, the verified correct option is No (square root is non-negative).

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
Write $x^{3/2}$ as a radical.

a. $x^{1/2} \cdot x$

b. $x\sqrt{x^2}$

c. $\sqrt{x^3}$ **or** $(\sqrt{x})^3$ ✓

d. $\sqrt[3]{x^2}$

Applying Radicals & Rational Exponents principles, $x^{3/2} = (x^{1/2})^3 = \sqrt{x^3}$. Therefore, the verified correct option is $\sqrt{x^3}$ or $(\sqrt{x})^3$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Find $f(x) = \sqrt{x} + \sqrt{1 - x}$. What is the domain?

a. $x \geq 0$

b. All reals

c. $0 < x < 1$

d. $0 \leq x \leq 1$ ✓

Applying Radicals & Rational Exponents principles, Need both $x \geq 0$ and $1 - x \geq 0$. Therefore, the verified correct option is $0 \leq x \leq 1$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Write $\sqrt[3]{a^2 b}$ using rational exponents.

a. $a^{2/3} b^{1/3}$ ✓

b. $a^{1/2} b^{1/3}$

c. $a^{3/2} b^{1/3}$

d. $a^2 b^3$

Applying Radicals & Rational Exponents principles, $(a^2 b)^{1/3} = a^{2/3} b^{1/3}$. Therefore, the verified correct option is $a^{2/3} b^{1/3}$.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Which of the following equals $8^{-2/3}$?

a. $\frac{1}{8}$

b. 4

c. $\frac{1}{4}$ ✓

d. -4

Applying Radicals & Rational Exponents principles, $8^{-2/3} = 1/(8^{2/3}) = 1/4$. Therefore, the verified correct option is $\frac{1}{4}$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the real number(s) equal to $(-32)^{3/5}$.

a. 8

b. Undefined

c. -8 ✓

d. -32^3

Applying Radicals & Rational Exponents principles, $(-32)^{1/5} = -2$. $(-2)^3 = -8$. Therefore, the verified correct option is -8.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Write $\sqrt[4]{x^3}$ using rational exponent.

a. x^{3+4}

b. $4x^3$

c. $x^{4/3}$

d. $x^{3/4}$ ✓

Applying Radicals & Rational Exponents principles, $\sqrt[n]{x^m} = x^{m/n}$. Therefore, the verified correct option is $x^{3/4}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify $\sqrt{4}$ — note: solve $\sqrt{36}$.

a. 9

b. 6 ✓

c. 3

d. 12

Applying Radicals & Rational Exponents principles, $\sqrt{4 \cdot 9} = 2 \cdot 3 = 6$. Therefore, the verified correct option is 6.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the number of real solutions of $\sqrt{x} = x^2 - 6x + 9$.

a. 0

b. 1

c. 4

d. 2 ✓

$\sqrt{x} = (x-3)^2$. Let $u = \sqrt{x}$: $u = (u^2 - 3)^2$. Solutions exist at $x = 0$ and $x = 4$ (verify graphically).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Polynomial Functions

Solve: $\sqrt{2x+3} = x$. Check for extraneous roots.

a. $x = 3$ **only** ✓

b. $x = -1$ only

c. $x = 3$ and $x = -1$

d. No solution

$2x + 3 = x^2 \Rightarrow x^2 - 2x - 3 = (x-3)(x+1) = 0$. $x = -1$: $\sqrt{1} = 1eq - 1$. Extraneous. $x = 3$: $\sqrt{9} = 3$. ✓.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Rational Expressions

Prove $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$ using rational exponents.

a. True only for $n = 2$

b. $(ab)^{1/n} = a^{1/n}b^{1/n}$ **— follows from power rules for real numbers** ✓

c. Not always true

d. Requires $a, b > 0$

Real number power rules: $(ab)^{1/n} = a^{1/n}b^{1/n}$ for $a, b \geq 0$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Simplify: $\frac{\sqrt{x+h}-\sqrt{x}}{h}$ (rationalize numerator).

a. $\frac{1}{2\sqrt{x}}$

b. $\frac{1}{\sqrt{x+h}+\sqrt{x}}$ ✓

c. $\frac{h}{\sqrt{x+h}+\sqrt{x}}$

d. $\frac{\sqrt{x+h}-\sqrt{x}}{h}$

Multiply by $\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

Express $\sqrt[6]{x^4}$ in simplest radical form.

a. $\sqrt[3]{x^2}$

b. $\sqrt[6]{x^4}$

c. $x^{2/3}$

d. **Both A and C ✓**

$x^{4/6} = x^{2/3} = \sqrt[3]{x^2}$. A and C are equal.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Conic Sections

Simplify $\frac{5+\sqrt{3}}{3-\sqrt{3}}$ by rationalizing.

a. $\frac{8+6\sqrt{3}}{6}$

b. $\frac{15+5\sqrt{3}+3\sqrt{3}+3}{6} = \frac{18+8\sqrt{3}}{6} = 3+\frac{4\sqrt{3}}{3}$ ✓

c. $5\sqrt{3}$

d. $\frac{5+\sqrt{3}}{3}$

Multiply by $(3+3)$: $(5+3)(3+3)9-3 = 18+836 = 3+433$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Trigonometric Functions

Find the real solutions of $x^{2/3} = 2$.

a. $x = n^{n+1/n}$ ✓

b. $x = n+1$

c. $x = n^2$

d. $x = n^{n/n+1}$

Raise both sides to $3/2$: $x = 2^{(3)/2}$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions & Their Graphs

Prove by induction that $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$. Step: assume true for n, verify for $n+1$.

a. Not provable by induction

b. Only for $n \le 100$

c. $\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$. ✓✓

d. Only for even n

Applying Sequences & Series principles, Standard induction step. Therefore, the verified correct option is $\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$. ✓.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Matrices & Systems

Determine the range of $f(x) = \frac{x}{x+1}$.

a. $0 \le y < 1$

b. $-1 < y < 1$

c. **All reals except** $y = 1$ ✓

d. All reals

As $x \to \infty$, $f \to 1$ (never equals 1). Solving $y = x/(x+1)$: $x = y/(1-y)$ defined for $yeq1$.

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